



Artificial Lifts and ESP design

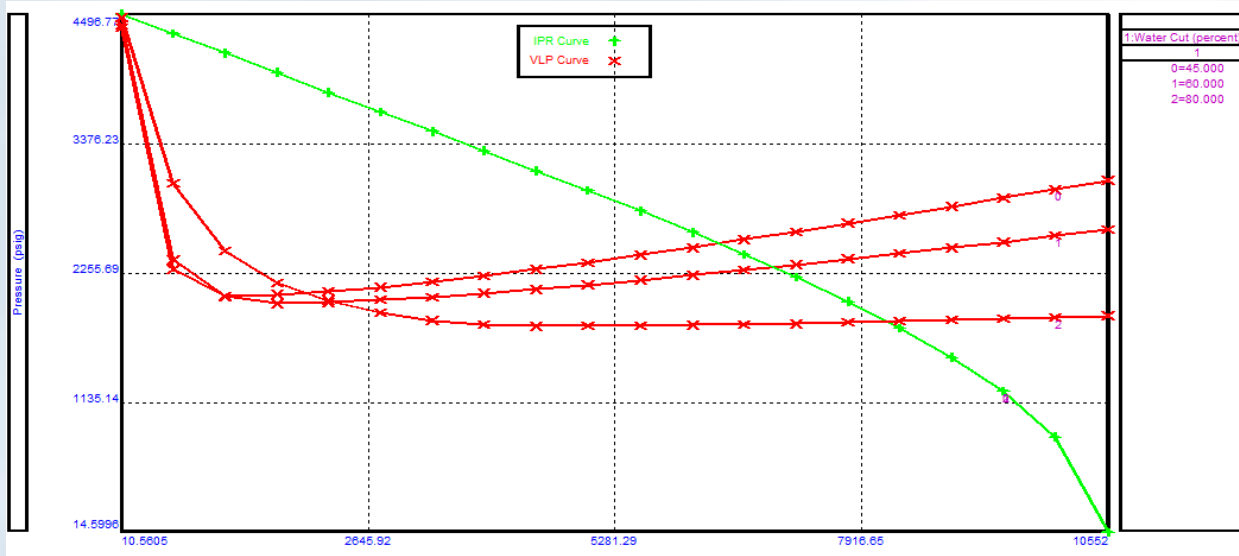
Outlines

- Systems' Sensitivity Analysis
- Introduction to Artificial Lift Systems and Comparisons
- Electrical Submersible Pump
 - Components
 - Important Variables and Plots to consider
 - Selection Criteria
- Exercise – ESP Design

Pressure loss and Sensitivity Analysis

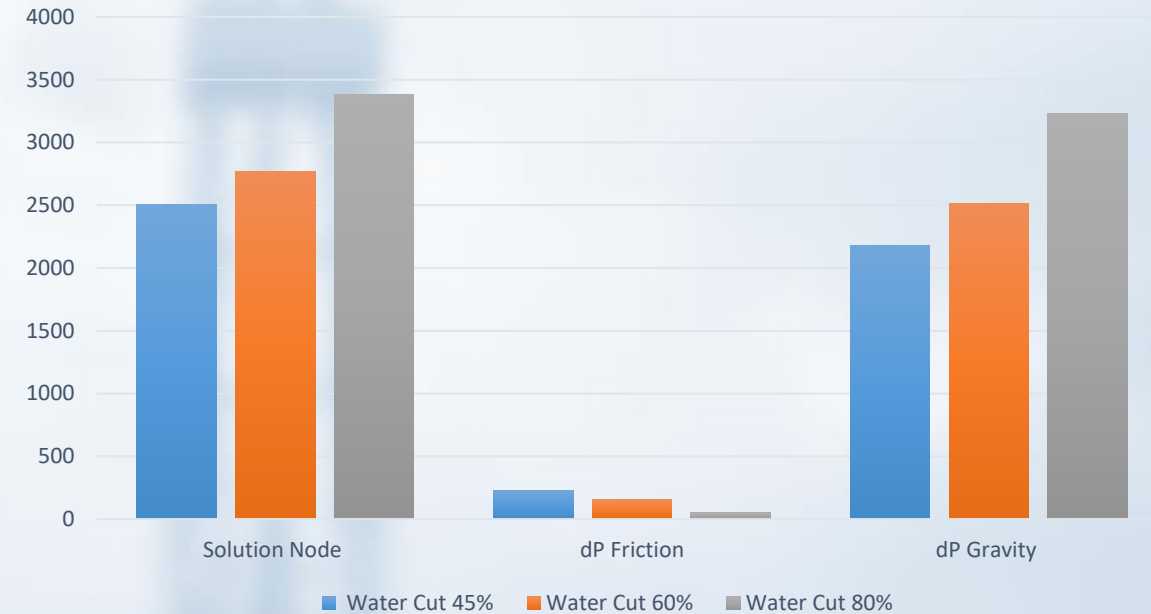
- The object of production increase is directly related to minimizing pressure drop in the wellbore as the fluid travels from bottom to the surface.
- Two variables which plays a big role in contributing to Pressure drop are:
 - Pressure loss due to gravity
 - Pressure loss due to friction
- Minimizing wellhead pressure also reduces to backpressure on the wellbore resulting in increase in production.
- System Sensitivity analysis evaluates the impact of changes in well equipment and reservoir fluids properties and quantifies the pressure drop associated with it.

Sensitivity Analysis – Water Cut



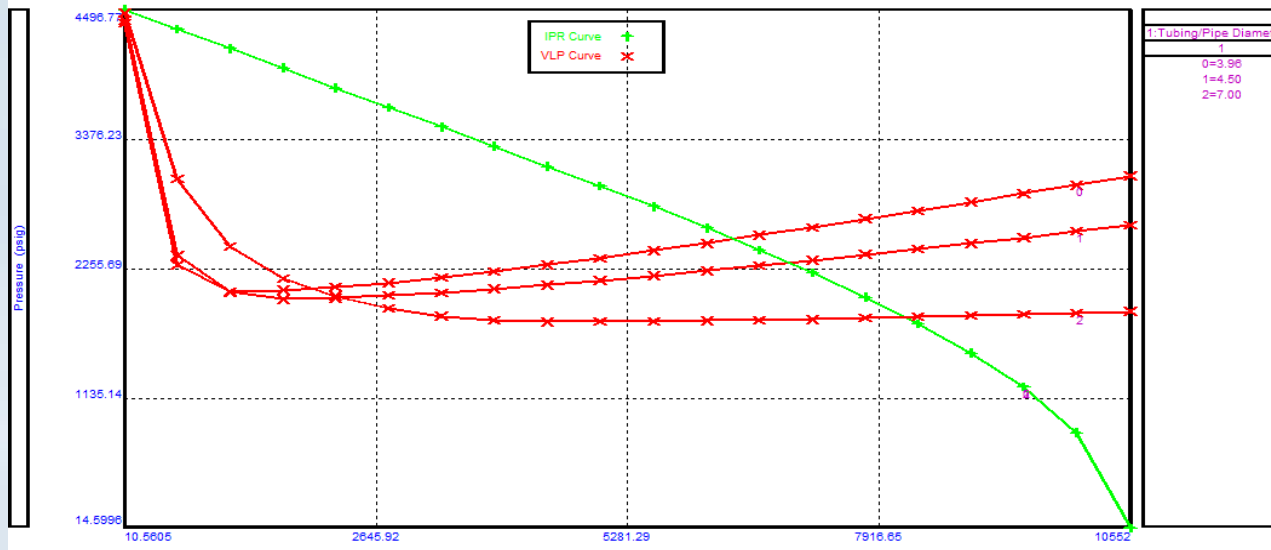
- Oil and fluid production increases with decrease in Water Cut %.
- Significant increase in Pressure loss with respect to water cut increase.
- Overall Pressure loss increases as dP gravity increases with Water Cut increase.

Sensitivity - Pressure Losses with different water cut



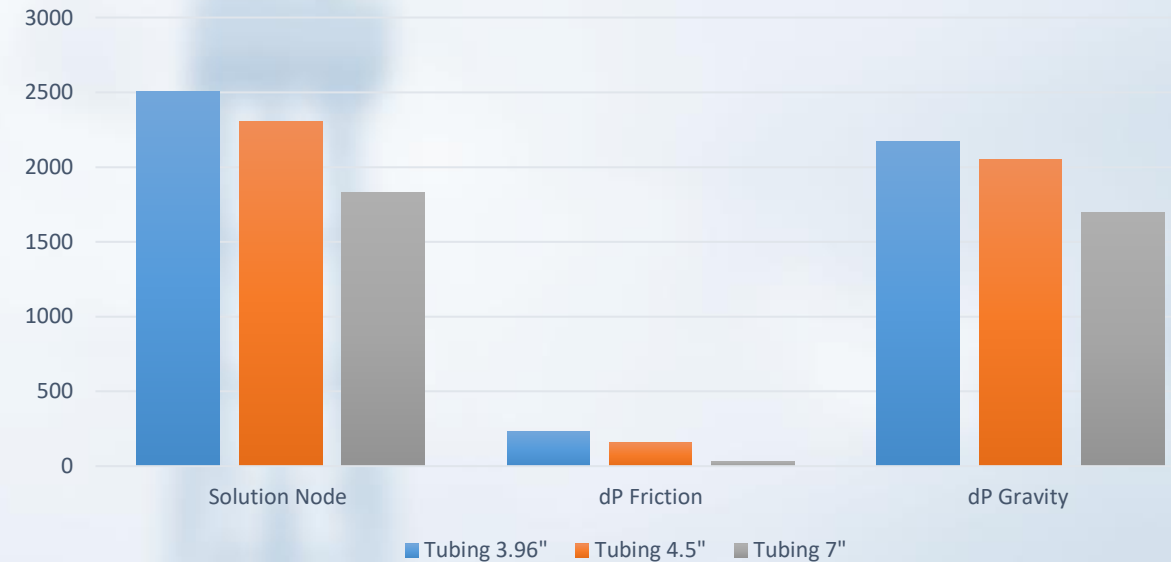
	Solution Node	dP Friction	dP Gravity	Oil Flowrate (bbl/d)
Water Cut 45%	2507	225	2176	3523
Water Cut 60%	2767.35	156	2510	2255
Water Cut 80%	3386	52	3236.5	728

System Analysis – Tubing Sizes



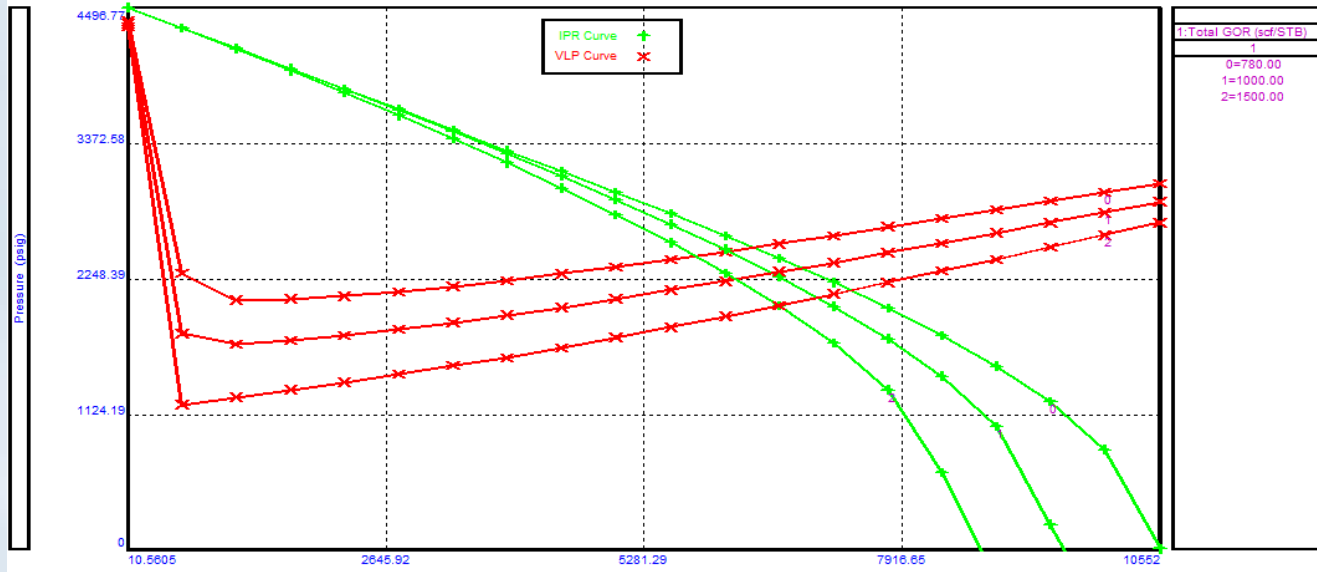
- Oil and fluid production increases with increase in Tubing ID.
- Pressure loss due to gravity and friction decreases as bigger Tubing ID is used.
- As overall Pressure loss decreases, the flow rate increases.

Sensitivity - Pressure Losses with different Tubing Size

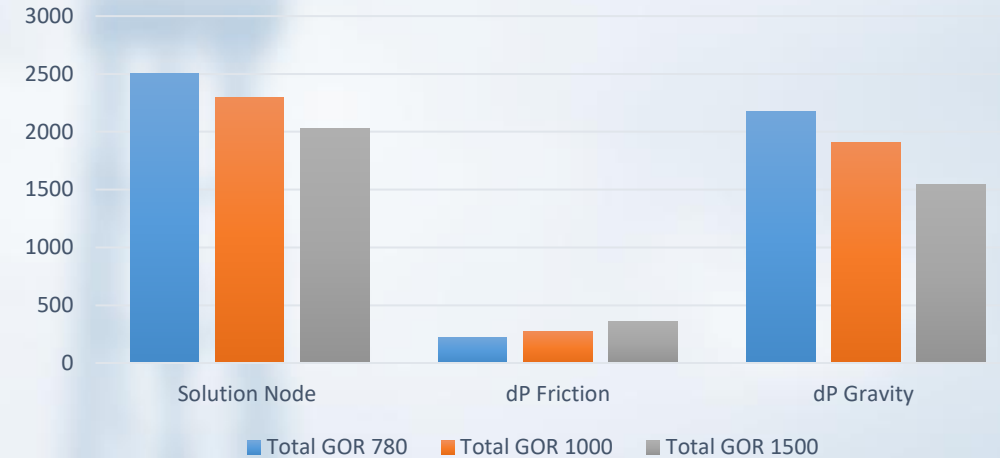


	Solution Node	dP Friction	dP Gravity	Oil Flowrate (bbl/d)
Tubing 3.96"	2509	230	2173	3520
Tubing 4.5"	2307	156	2049	3836
Tubing 7"	1833	31	1697	4508

Sensitivity Analysis – GOR



Sensitivity - Pressure Losses with different GOR

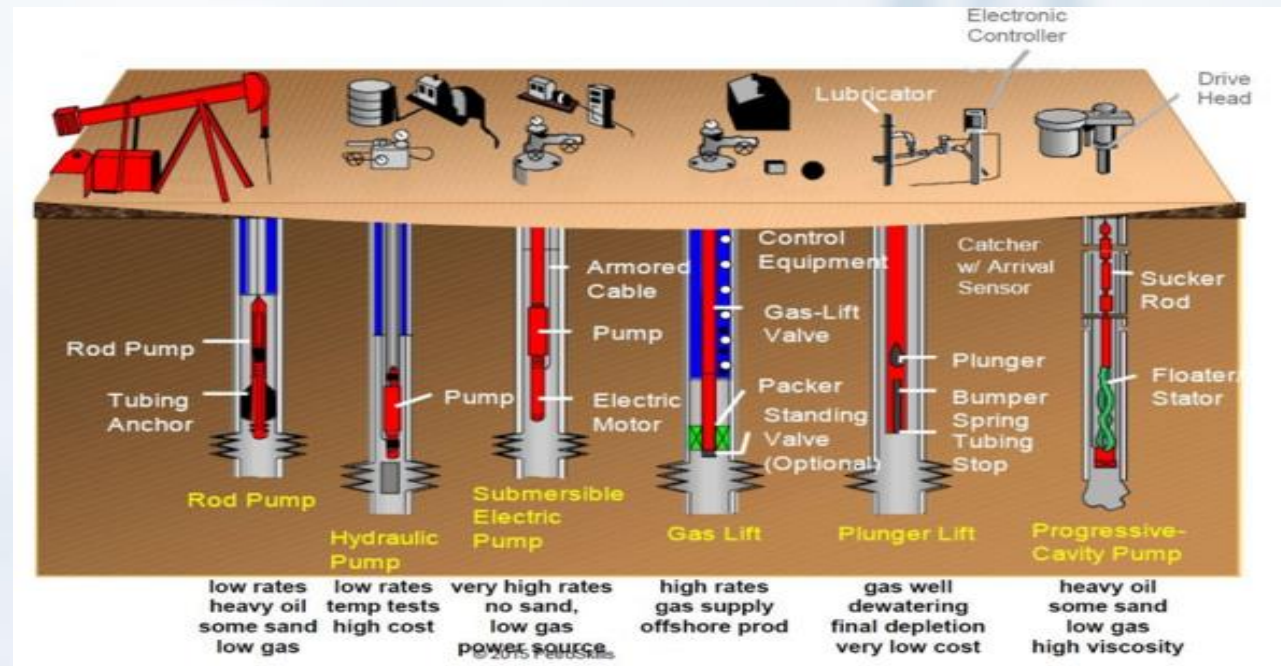


- Oil and fluid production increases with increase in GOR.
- Pressure loss due to gravity decreases with increase in GOR and pressure loss due to friction increases.
- Overall Pressure loss decreases as dP gravity outweighs dP friction as the Total GOR increases.

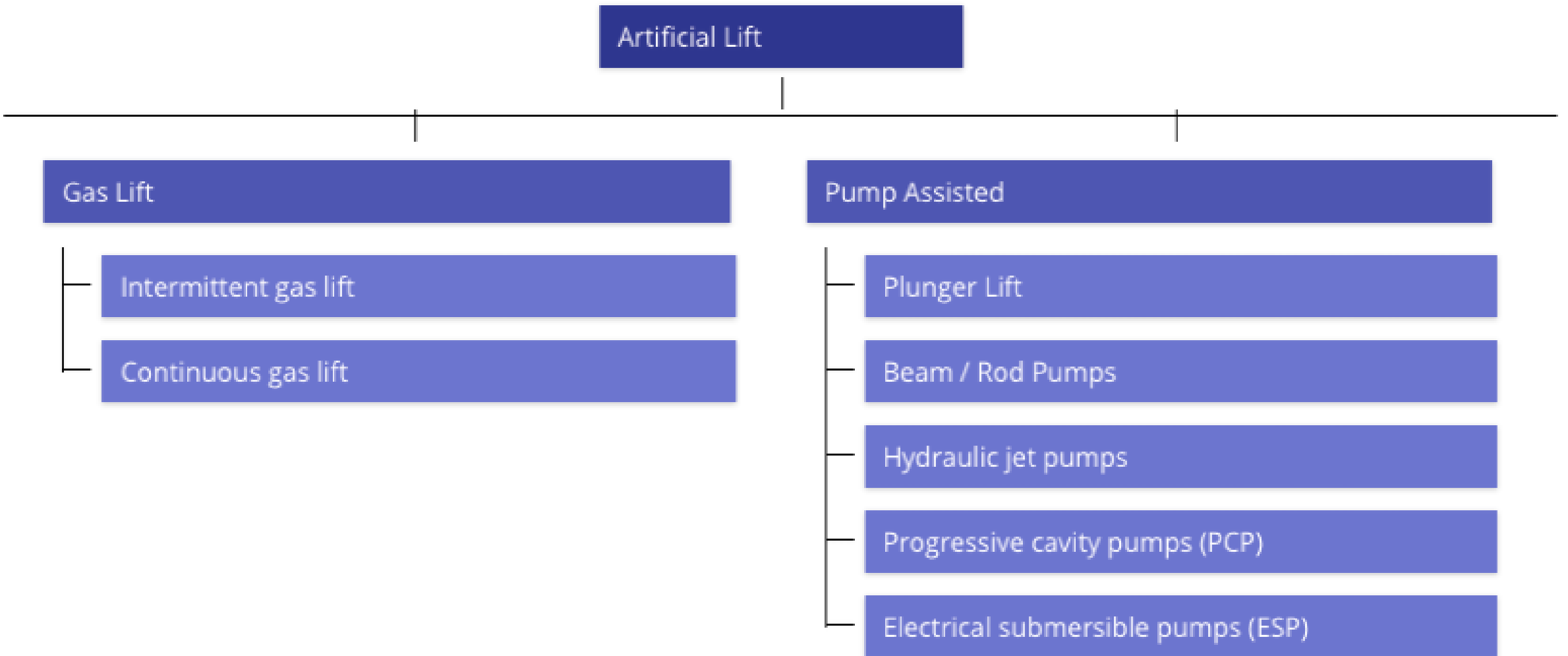
	Solution Node	dP Friction	dP Gravity	Oil Flowrate (bbl/d)
Total GOR 780	2507.6	225.6	2176.7	3523
Total GOR 1000	2295	275	1908	3626
Total GOR 1500	2028	359	1546	3664

Artificial lift Systems

- Artificial lift is a method used to lower the producing bottom hole pressure (P_{wf}) on the formation to obtain a higher production rate from the well.
- Artificial lift can be used to generate flow from a well in which no flow is occurring or used to increase the flow from a well to produce at a higher rate.
- Most oil wells require artificial lift at some point in the life of the field, and many gas wells benefit from artificial lift to take liquids off the formation so gas can flow at a higher rate.



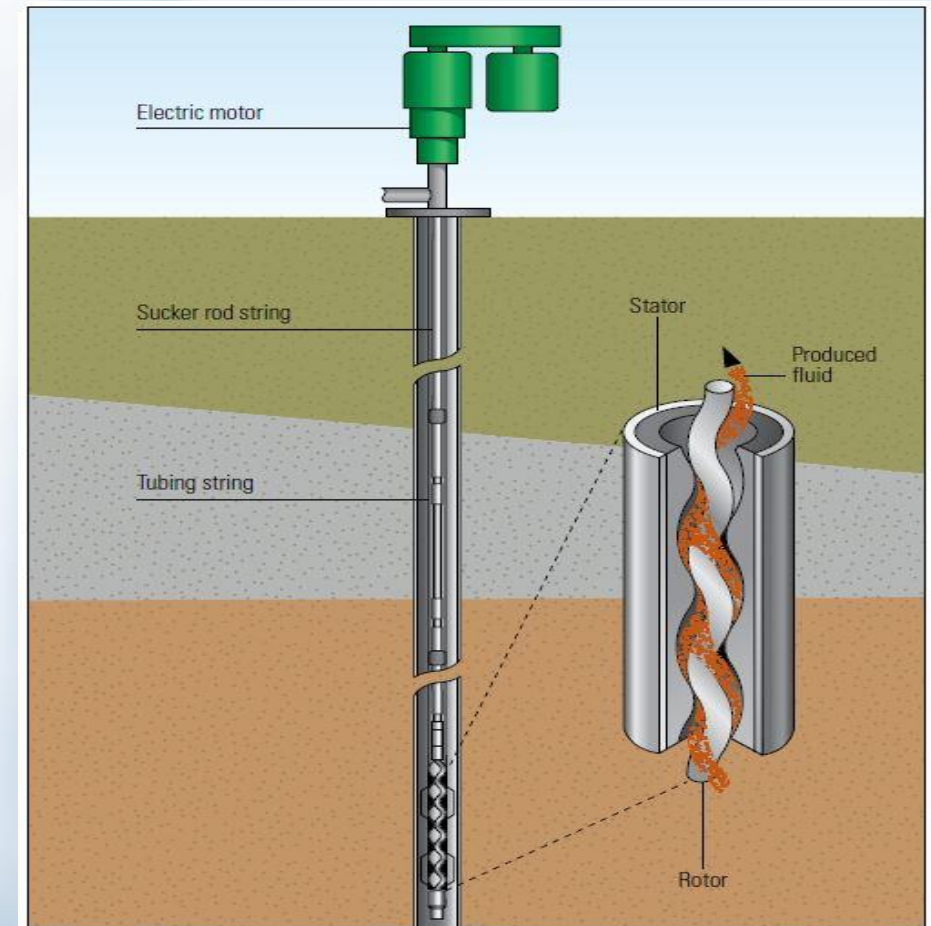
Artificial Lift Types



Positive Displacement Pumps

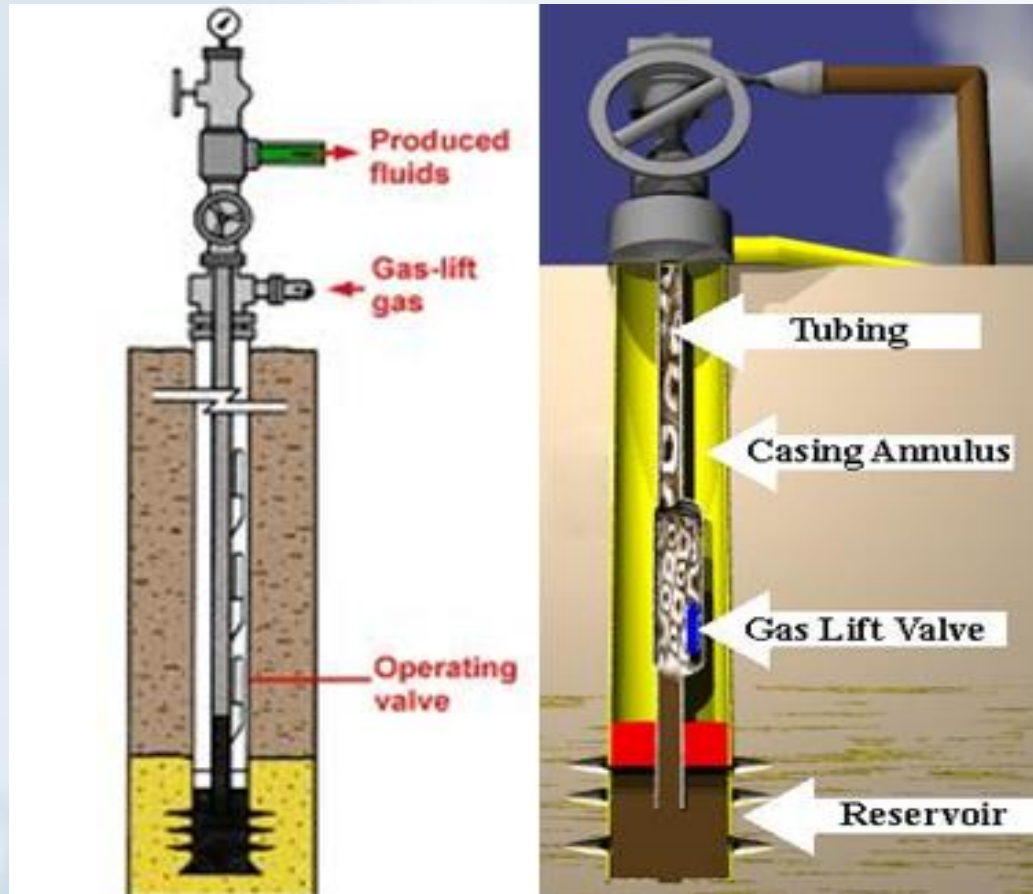


Sucker Rod Pump (Beam Pump)

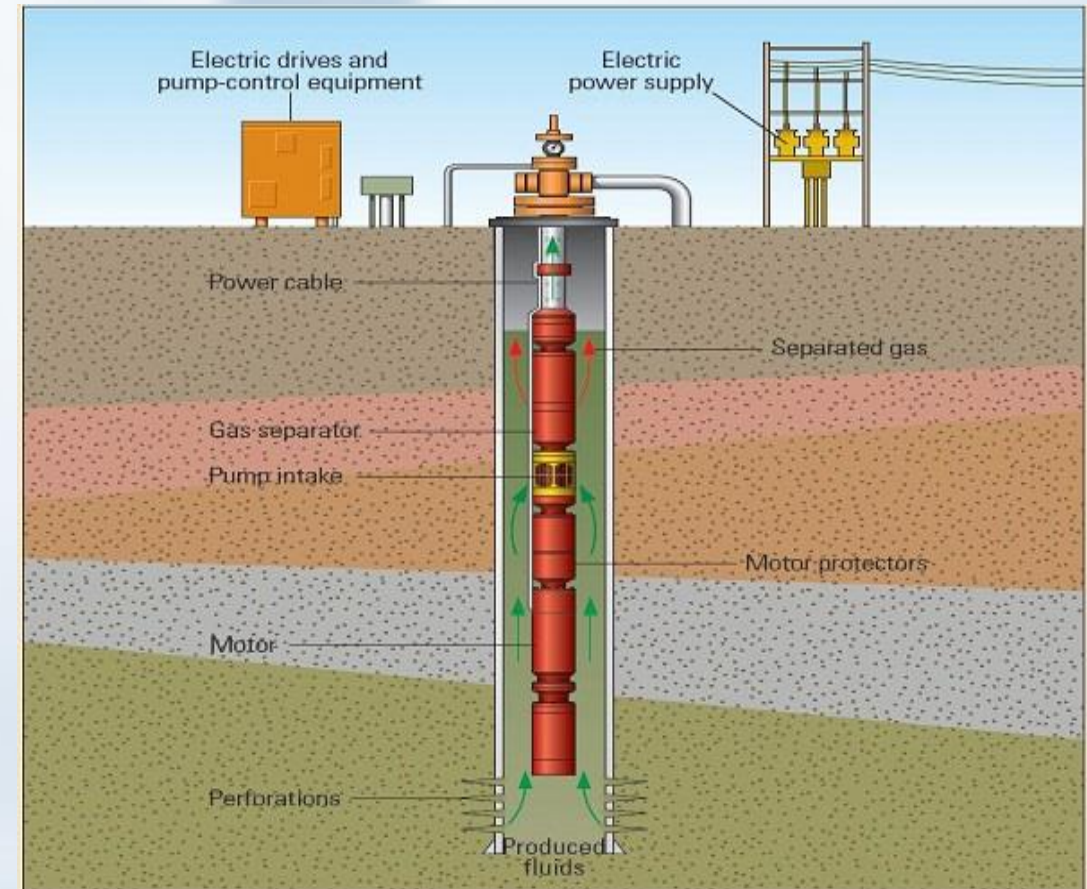


Progressive Cavity Pump

- Gas Lifts



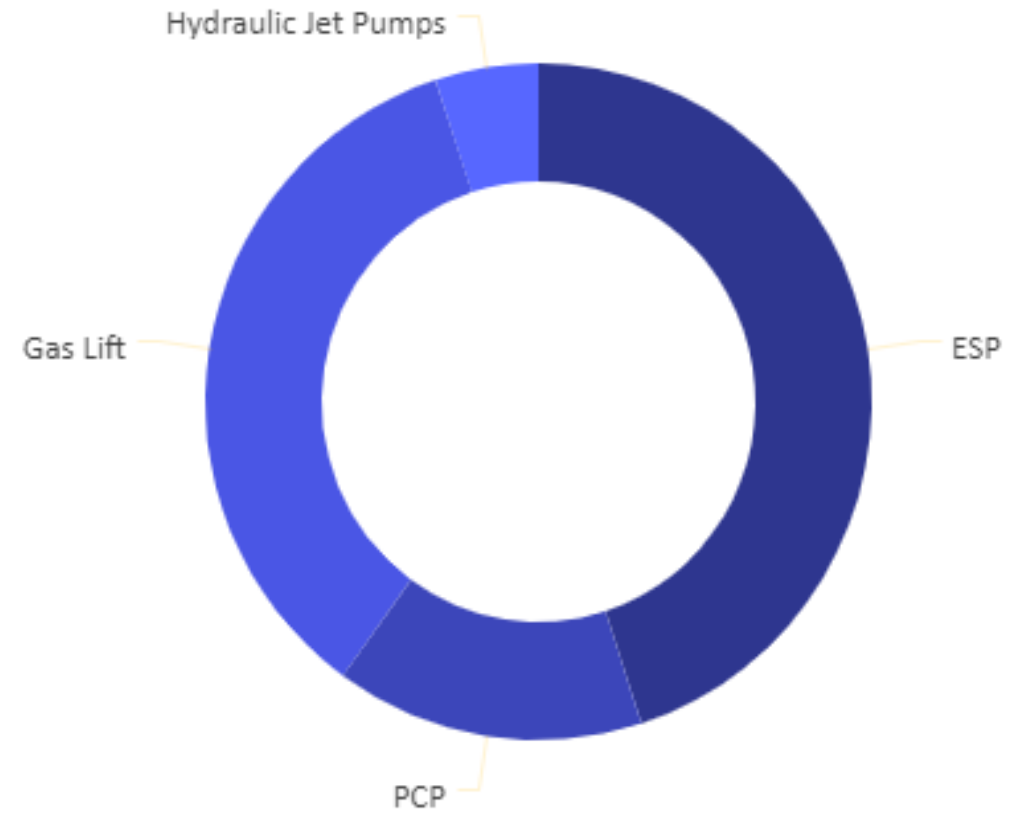
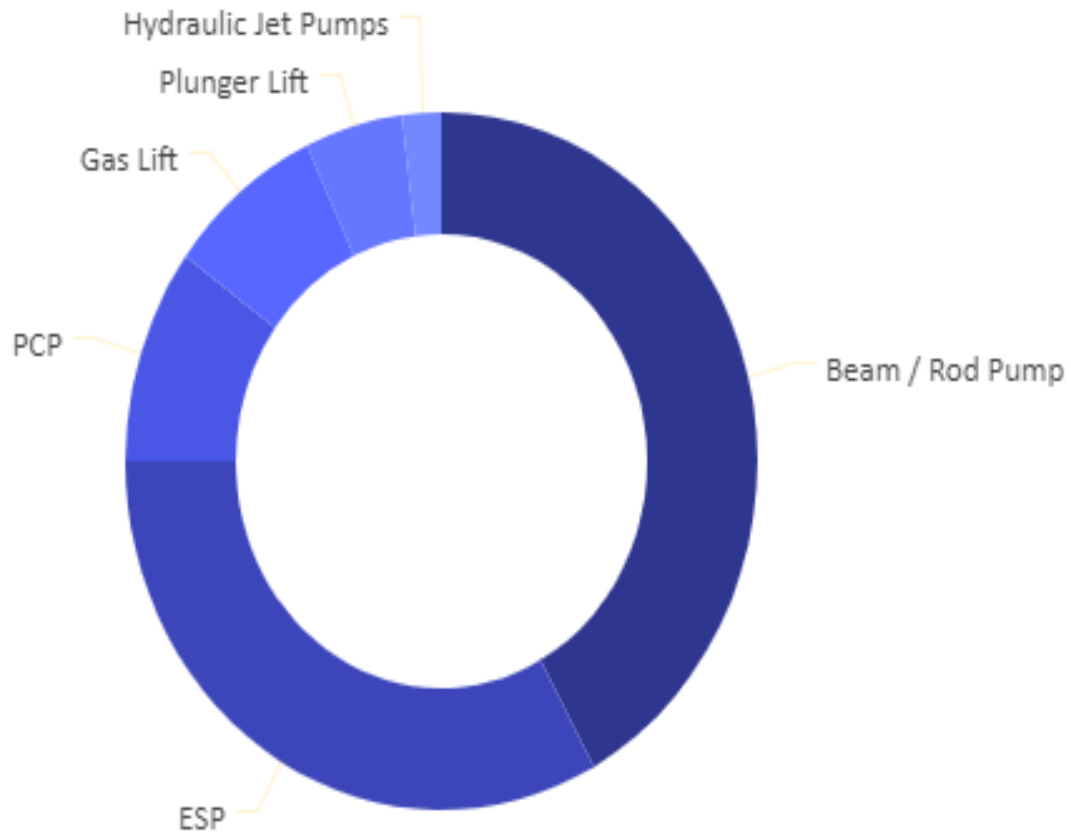
- Electrical submersible Pump



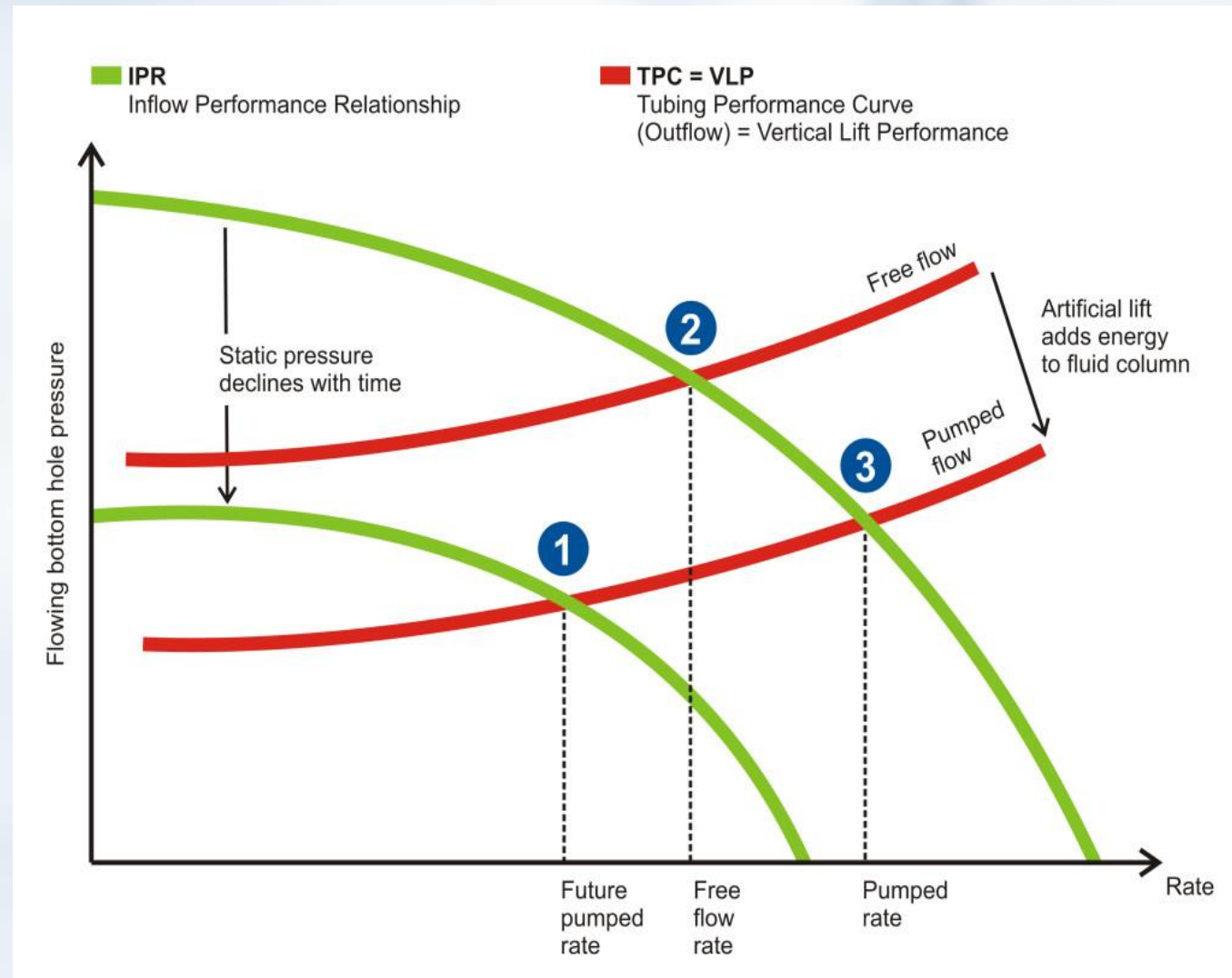
Market of Artificial lifts by type

- Global Breakdown by Technology

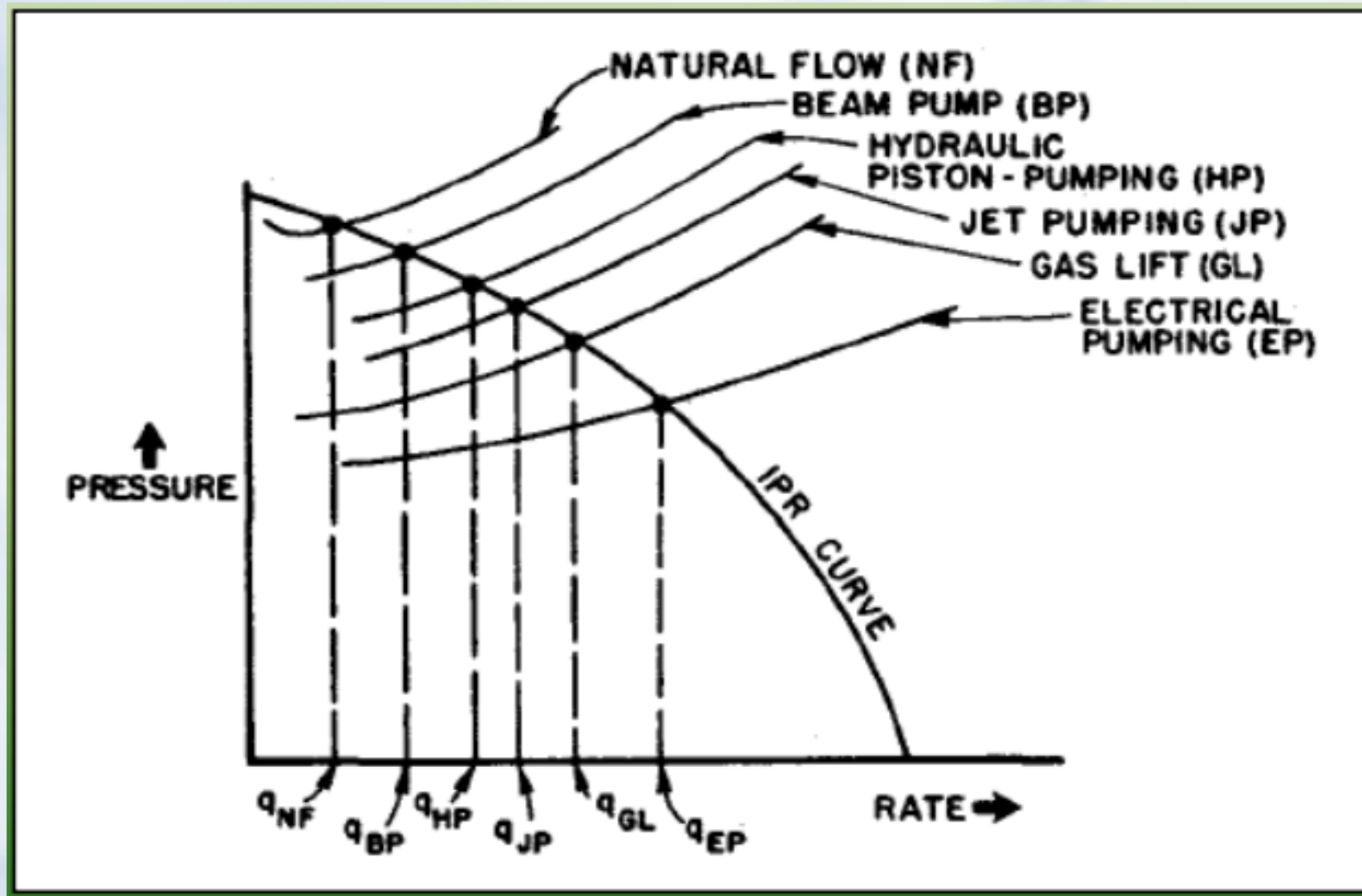
- Global breakdown in offshore



Production enhancement through Artificial lift



Effects of artificial lifts on production system



Artificial Lift Selection Criteria – (Weatherford 2005)

Operating Parameters	Positive displacement pumps			Dynamic displacement pumps		Gas lift	Plunger lift
	<i>Rod pump</i>	<i>PCP</i>	<i>Hydraulic Piston</i>	<i>ESP</i>	<i>Hydraulic Jet</i>		
Typical Operating Depth (TVD)	100 to 11000 ft	2000 to 4500 ft	7500 to 10000 ft		5000 to 10000 ft	5000 to 10000 ft	To 8000 ft
Maximum Operating Depth (TVD)	16000 ft	6000 ft	17000 ft	15000 ft	15000 ft	15000 ft	20000 ft
Typical Operating Volume	5 to 1500 BFPD	5 to 2200 BFPD	50 - 500 BFPD	100 to 30000 BFPD	300 - 4000 BFPD	100 - 10000 BFPD	1 - 5 BFPD
Maximum	6000 BFPD	4500	4000 BFPD	40000	>15000	30000	200 BFPD

Artificial Lift Selection Criteria – (Weatherford 2005)

Operating Parameters	Positive displacement pumps			Dynamic displacement pumps		Gas lift	Plunger lift
	<i>Rod pump</i>	<i>PCP</i>	<i>Hydraulic Piston</i>	<i>ESP</i>	<i>Hydraulic Jet</i>		
Operating Volume		BFPD		BFPD	BFPD	BFPD	
Typical Operating Temperature	100 - 350 ° F [40-177 ° C]	75 - 150 ° F [24-65 ° C]	100 - 250 ° F [40-120 ° C]		100 - 250 ° F [40-120 ° C]	100 - 250 ° F [40-120 ° C]	120 ° F [50 ° C]
Maximum Operating temperature	550 ° F [288 ° C]	250 ° F [120 ° C]	500 ° F [260 ° C]	400 ° F [205 ° C]	500 ° F [260 ° C]	400 ° F [205 ° C]	500 ° F [260 ° C]
Typical Wellbore Deviation	0 - 20 deg landed pump	N/A	0 - 20 deg landed pump		0 - 20 deg hole angle	0 - 50 deg	N/A
Maximum Wellbore Deviation	0 - 90 deg landed pump	0 - 90 deg < 15 deg/100 ft	0 - 90 deg < 15 deg/100 ft	0 - 90 deg	0 - 90 deg < 24 deg/100 ft	70 deg, short to medium radius	80 deg

Artificial Lift Selection Criteria – (Weatherford 2005)

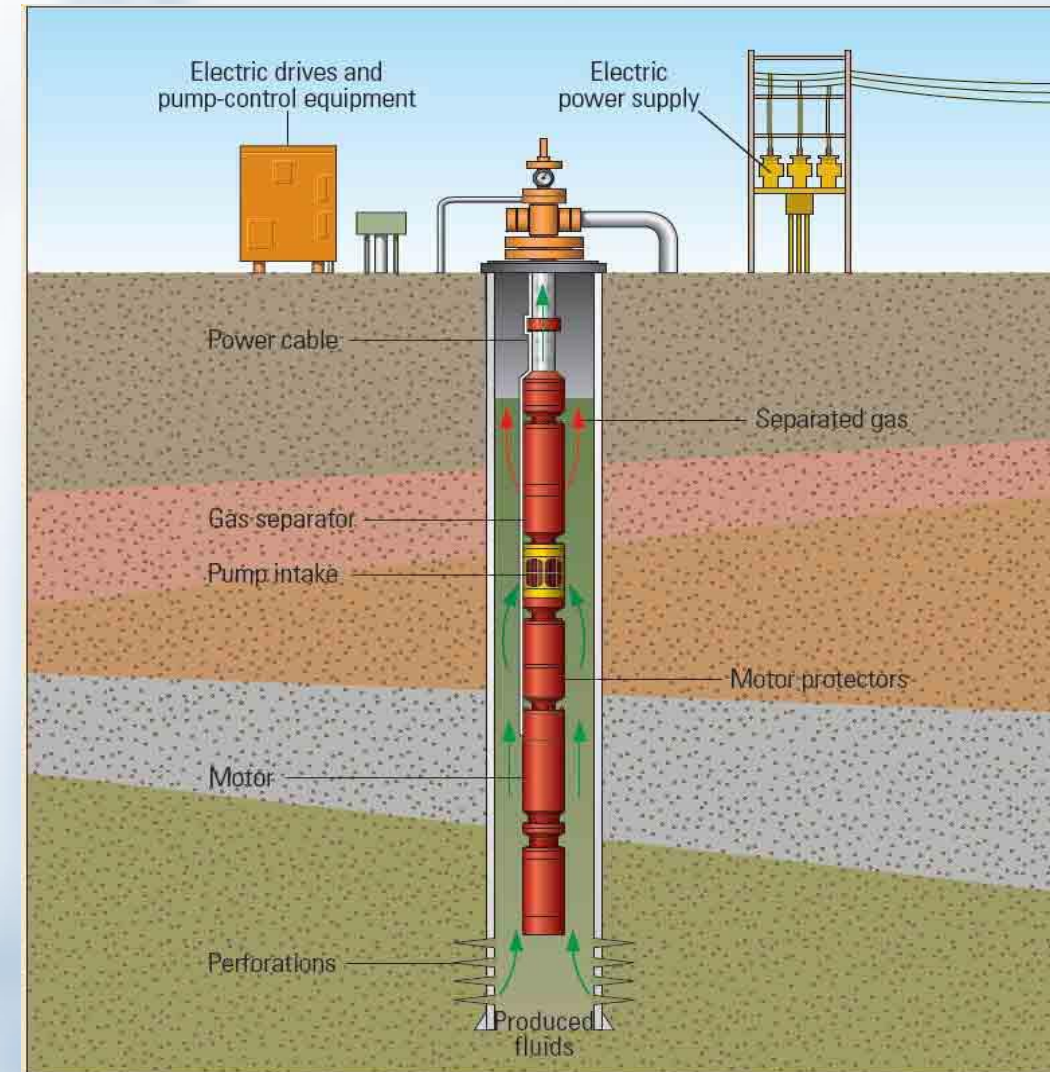
Operating Parameters	Positive displacement pumps			Dynamic displacement pumps		Gas lift	Plunger lift
	<i>Rod pump</i>	<i>PCP</i>	<i>Hydraulic Piston</i>	<i>ESP</i>	<i>Hydraulic Jet</i>		
Operating Volume		BFPD		BFPD	BFPD	BFPD	
Corrosion handling	Good to Excellent	Fair	Good	Good	Excellent	Good to excellent	Excellent
Gas handling	Fair to good	Good	Fair	Fair	Good	Excellent	Excellent
Solids handling	Fair to good	Excellent	Poor	Fair	Good	Good	Poor to Fair
Fluid gravity	> 8 ° API	< 35 ° API	> 8 ° API	> 10 ° API	> 8 ° API	> 15 ° API	GLR = 300 SCF/Bbl per 1000 ft of depth
Servicing	<u>Workover or pulling rig</u>	<u>Workover or pulling rig</u>	<u>Hydraulic or wireline</u>	<u>Workover or pulling rig</u>	<u>Hydraulic or wireline</u>	<u>Wireline or workover rig</u>	<u>Wellhead catcher or wireline</u>
Prime mover	Gas or electric	Gas or electric	Multi-cylinder or electric	Electric motor	Multi-cylinder or electric	Compressor	Well's natural energy
Offshore applications	Limited	Good	Good	Excellent	Excellent	Excellent	N/A
System efficiency	45% - 60%	40% - 70%	45% - 55%	35% - 60%	10% - 30%	10% - 30%	N/A

Electrical Submersible Pump (ESPs)

The electrical submersible pump, typically called an ESP, is an efficient and reliable artificial-lift method for lifting moderate to high volumes of fluids from wellbores. These volumes range from a low of 150 B/D to as much as 150,000 B/D (24 to 24,600 m³/d). Variable-speed controllers can extend this range significantly, both on the high and low side. The ESP's main components include:

- Multistage centrifugal pump
- Three-phase induction motor
- Seal-chamber section
- Power cable
- Surface controls

The components are normally tubing hung from the wellhead with the pump on top and the motor attached below. There are special components in which this configuration is inverted



ESP – Downhole Equipment

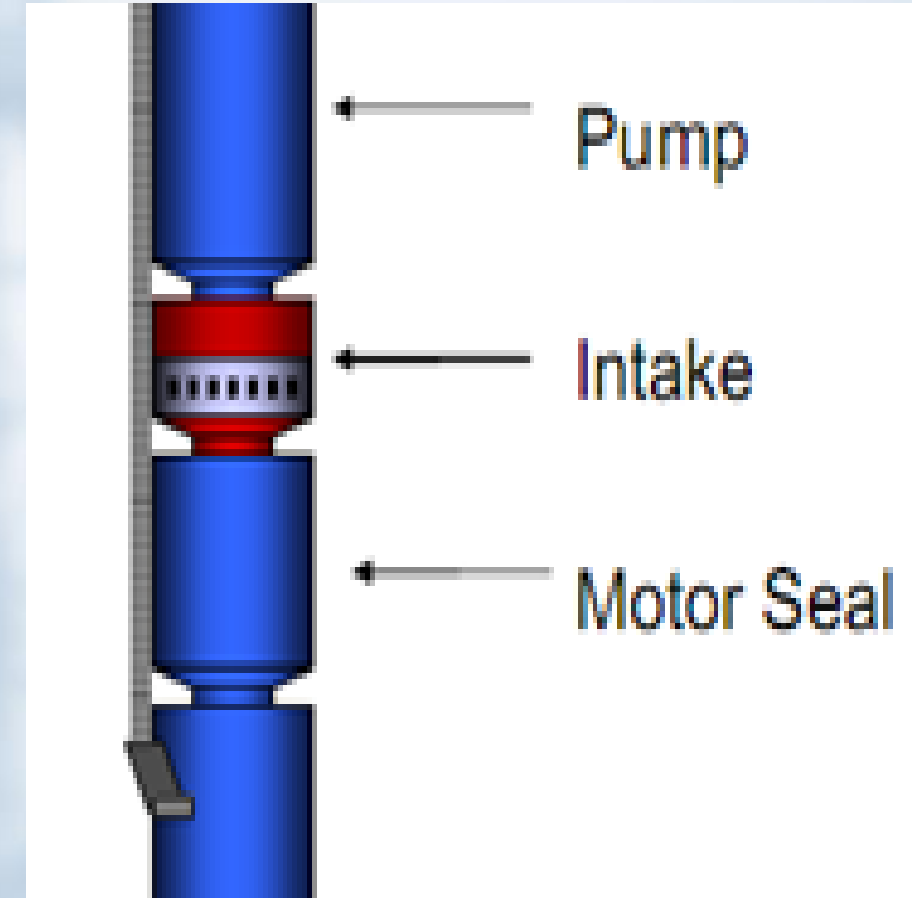
Pump Intake

- As the name suggests “ Pump Intake ” is where the well fluid enters the Submersible Pumping System.
- Care should be taken when designing a submersible pump intake because it is such a vital point in the system
- The amount of free gas by volume at pump intake conditions should be no more than 10% to 15% by volume.

Motor Seal

Installed below the intake and above the electric motor. Also called Protector

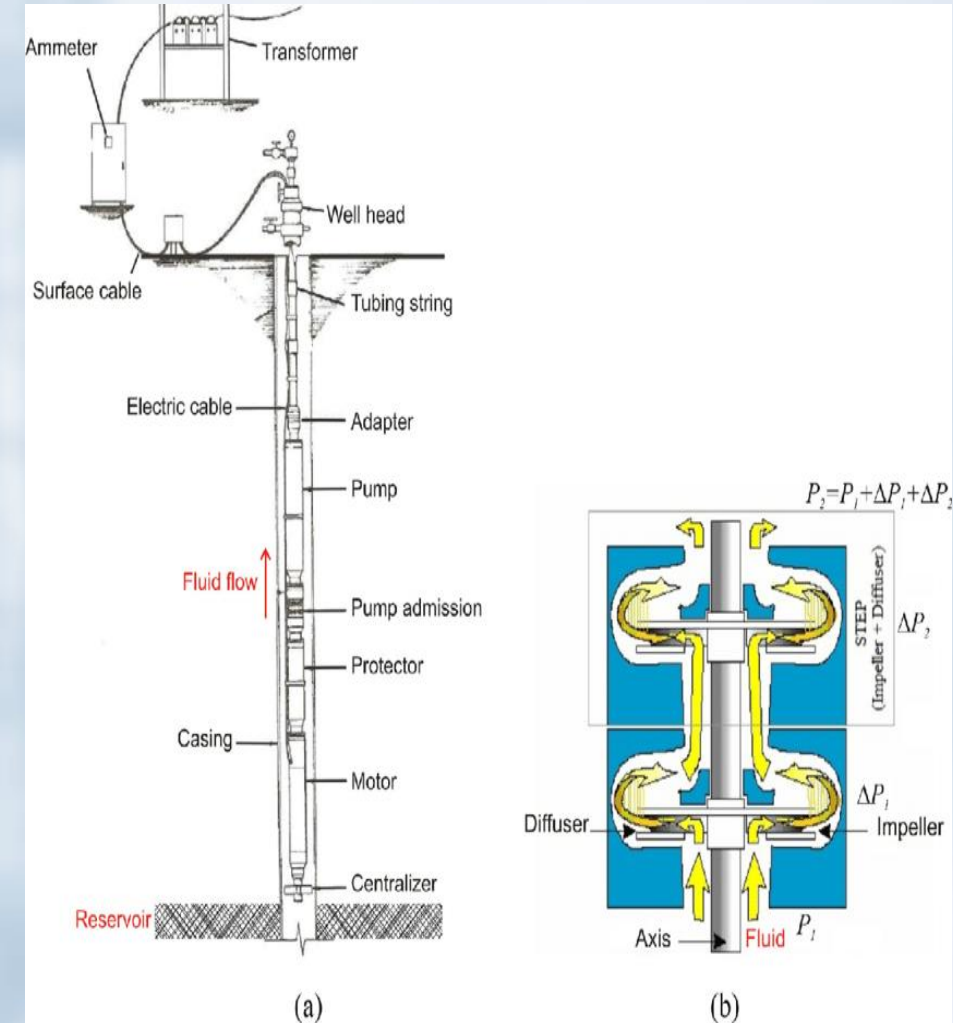
- Equalizes the pressure in the wellbore with the pressure inside the motor,
- Provides area for motor oil expansion volume (induced by temperature changes in the motor),
- Isolates the well fluid from the clean motor oil,
- Absorbs the pump shaft thrust load



ESP – Downhole Equipment

Centrifugal Pump

- The centrifugal pump is a multi-stage pump, containing a selected number of impellers equipped with vanes, inside a closely fitted diffuser, located in series on an axial shaft, driven by the electrical motor.
- It creates pressure by the rotation of a series of vanes in an impeller. The motion of the impeller forms a partial vacuum at the suction end of the impeller.
- The impeller's job is to transfer energy by rotation to the liquid passing through it, thus raising the kinetic energy. The diffuser section then converts this energy to potential energy, raising the discharge pressure.
- Outside forces, such as the atmospheric pressure or weight of a column of liquids, push fluid into the impeller eye and out to the periphery of the impeller. From there, the rotation of the high-speed impeller throws the liquid into the diffuser.
- Each “pump stage” consists of an impeller and a diffuser. Again, the impeller takes the fluid and imparts kinetic energy to it. The diffuser converts this kinetic energy into potential energy (head).



ESP – Downhole Equipment

Motor

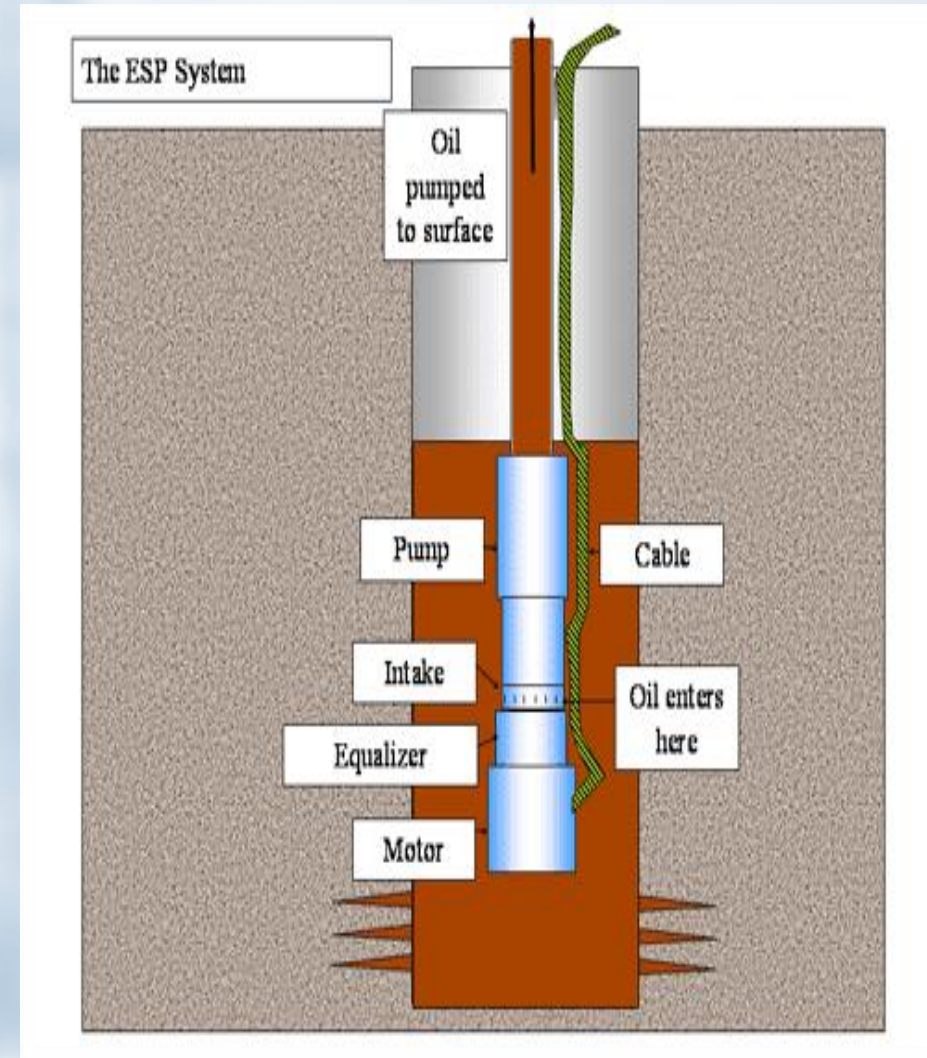
ESP motor is installed below the motor seal and above the downhole sensor. In cases where a downhole sensor is not installed, the motor is installed at the very bottom of ESP string, generally attached to a motor guide.

ESP motor is an induction motor, two poles, three phases, squirrel cage type stator winding filled with specific motor oil, high dielectric strength (> 28 KV).

The motor is rated for a specific horsepower, voltage, & current. Its role is to drive the downhole pump and seal section

Motor Cable

The main cable consists of three conductor wires extending from the top of the motor flat lead extension to the wellhead banded to the production tubing. The ESP cable carries current (amperage) from the motor controller at the surface down to the motor



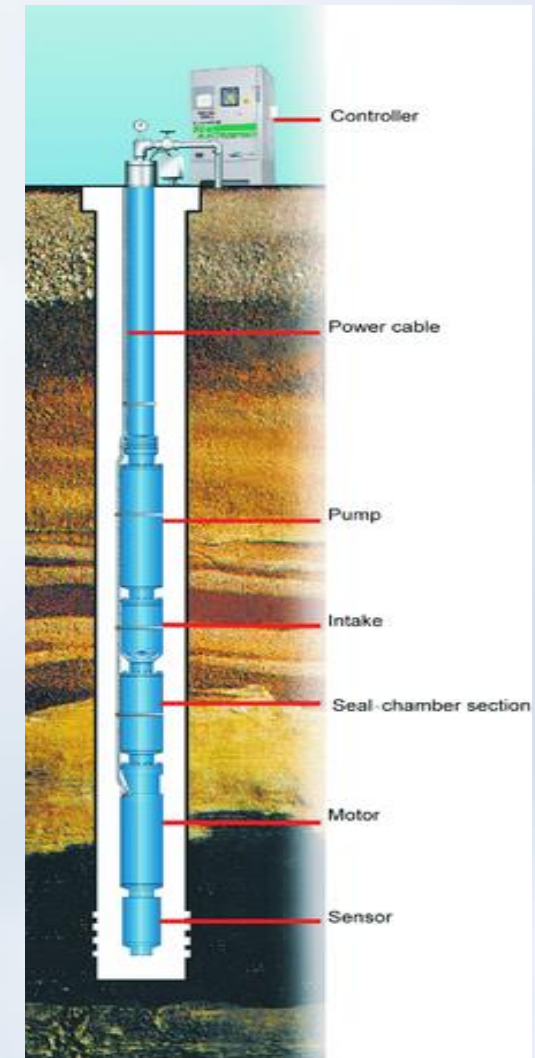
ESP – Surface Equipment

Transformer

- The transformer is used to convert the incoming voltage at the location to the correct voltage for the ESP motor.

Motor Controllers

- The surface controller provides power to the ESP motors and protects the downhole ESP components. This page discusses the features of these controllers.
- There are three types of motor controllers used on ESP applications and all are generally specifically designed for application with ESPs. They include the switchboard, soft starter, and the variable speed controller.
- All units vary in design, physical size, and power ratings.



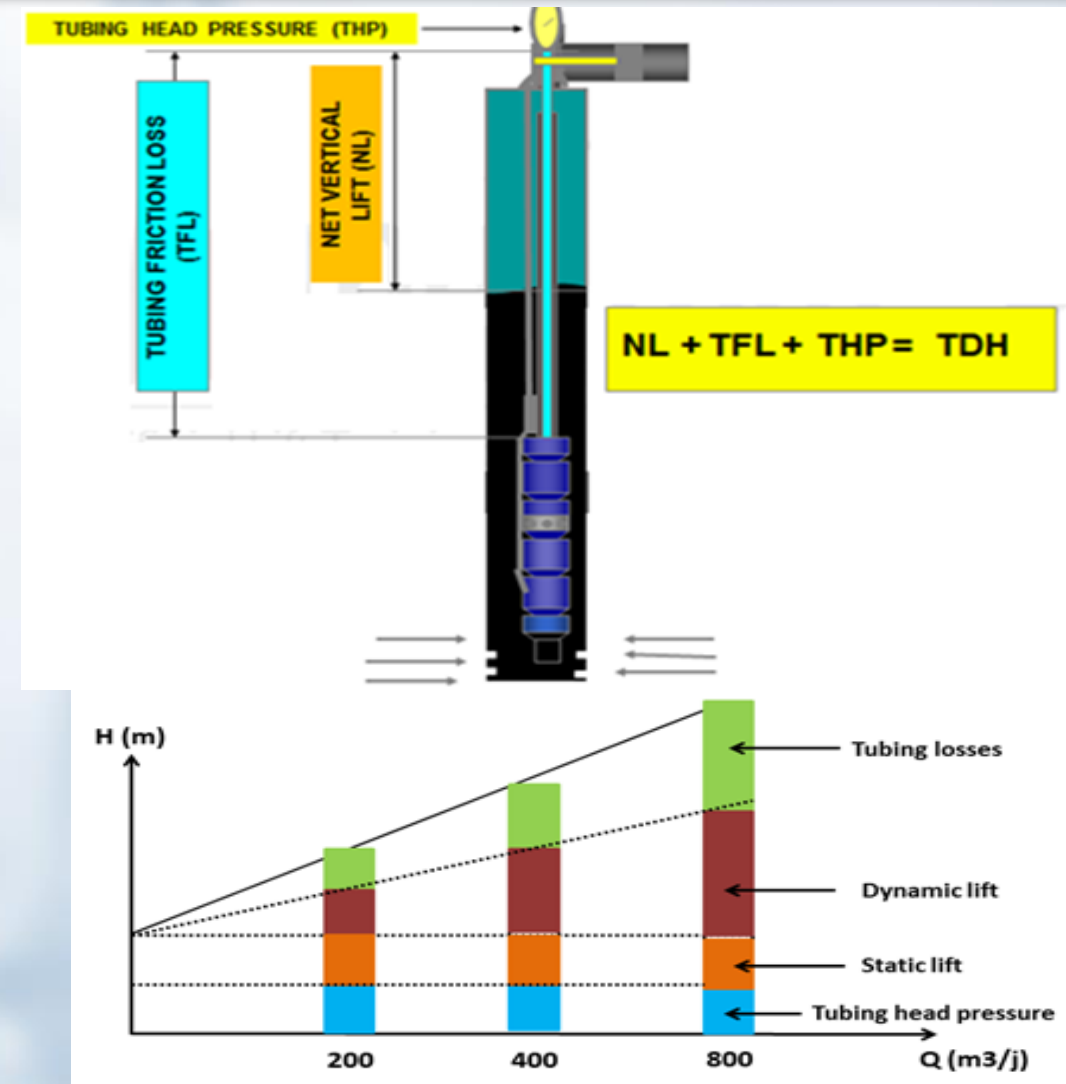
Total Dynamic Head

TDH is the sum of three basic components:

- Net Vertical Lift (NL) = is the net distance where the fluid must be lifted,
- Tubing Friction Loss (TFL) = Flow disturbance in the tubing string during pumping process,
- Tubing Head Pressure (THP) = Pressure which the unit must pump against (back pressure caused by choking on well head).

Total Dynamic Head = Net Vertical Lift + Tubing Friction Loss + Tubing Head Pressure

After calculating TDH, subsequent calculation needs to be done to get the number of the stages and motor HP required.



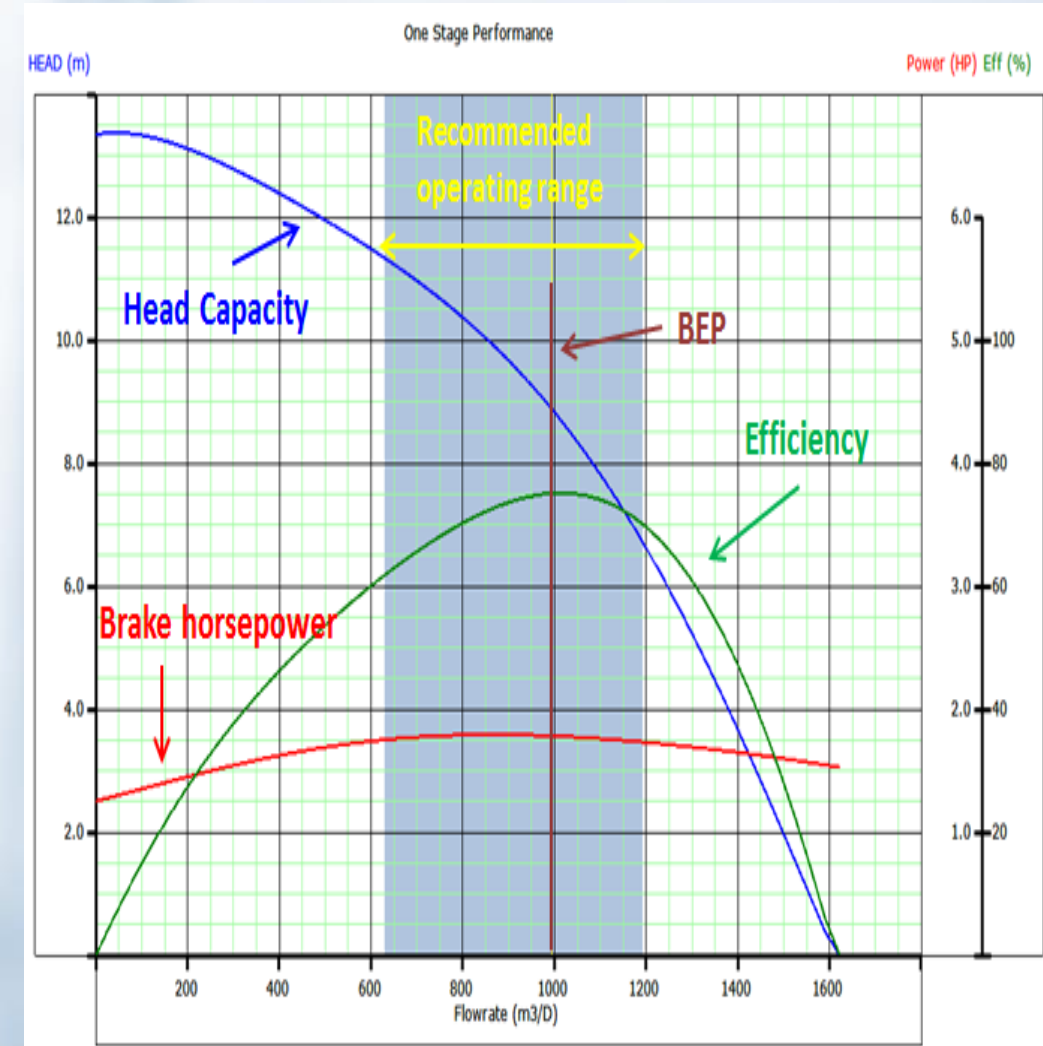
Pump Performance Curves

- The left vertical axis is scaled in feet and meters of head (or lift).
- The bottom horizontal axis is scaled in bbl/d and m³/d.
- The curve labeled Head-Capacity defines the lift (or head) the impeller can produce at all of the available flow rates.
- The first vertical axis on the right is scaled in horsepower. It is based on pumping water with a specific gravity of 1.00

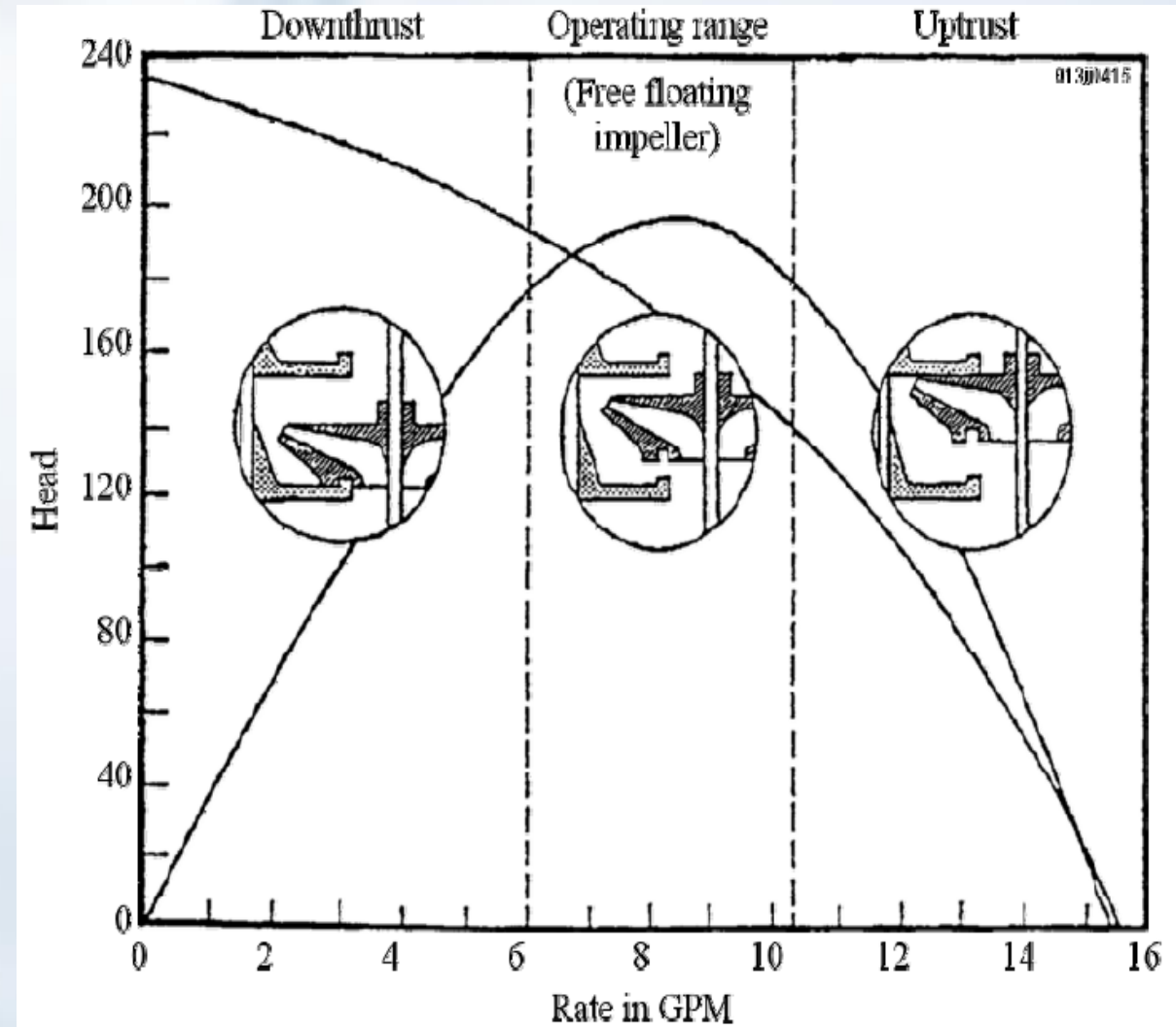
Brake horsepower (BHP): the power required by the pump corrected for a fluid with a S.G. = 1.

Best Efficiency Point (BEP): Defines pump performance parameters at the maximum value on the efficiency curve.

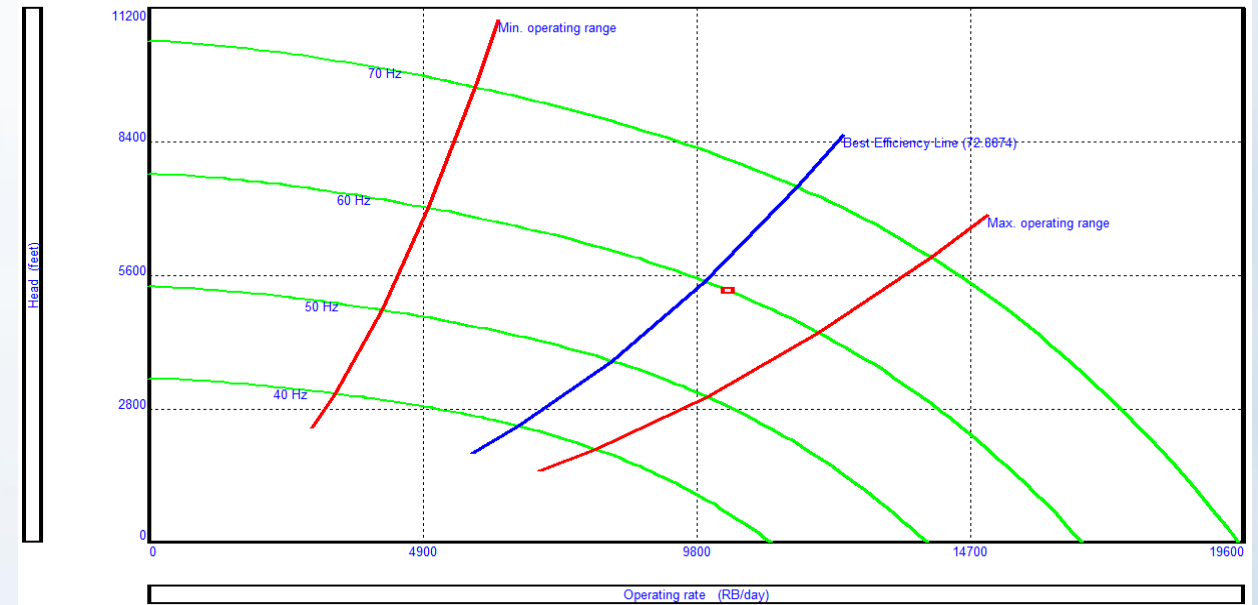
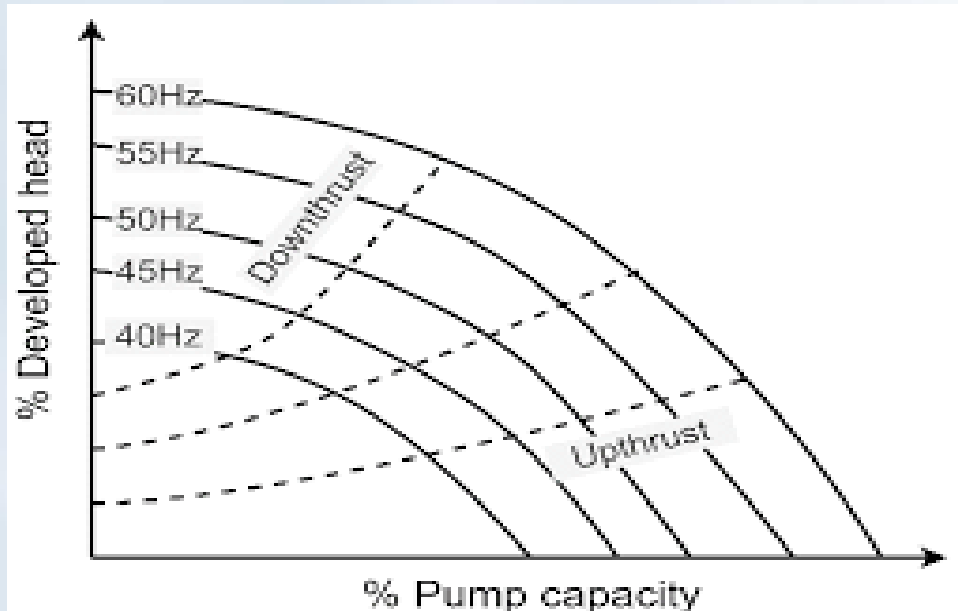
Recommended operating range: located between the maximum and minimum recommended flow rates.



- All centrifugal pumps develop axial thrust.
- Axial thrust is an important factor in establishing the recommended operating range of a centrifugal pump.
- Refer to following Figure, which depicts a typical thrust curve vs. capacity over the entire range of a particular pump type. It shows where up-thrust and down-thrust occurs in relation to the head capacity curve.



Design Evaluation – Pump Plot



- More Power (Horse Power) can generate more lift but economics plays a big factor.
- Simulation software like Prosper, Pipesim, etc can depict the pump plot based on the various inputs and required head.
- The right hand side plot shows different Operating curves on the a operating rate vs head plot and the red dot close to right hand side of the Best efficiency line which is desirable.

Design and Installation Considerations for ESPs

Categories	Considerations
Capital Costs	- Relatively low capital cost if electric power available. - Costs increase as horsepower increases.
Downhole equipment	- Requires proper cable in addition to motor, pumps, seals, etc. - Good design plus good operating practices are essential.
Miscellaneous problems	- Requires a highly reliable electric power system. - System very sensitive to changes downhole or in fluid properties requires a highly reliable electric power system. - System very sensitive to changes downhole or in fluid properties
Casing Size Limit	- Casing size will limit use of large motors and pumps. - Avoid 4.5" casing and smaller. - Reduced performance inside 5.5" casing depending on depth and rate
Operating Costs	- Varies. If high HP, high energy costs.
Solid Handling Capacities	- Poor. Requires <100 – 200 PPM solids for standard construction.
Gas Handling Capacities	- Poor for free gas (i.e. >5% through pump).
Temperature limitation	- Limited to <250°F for standard & <400°F for special motors & cable. - 100 – 275°F typical.

Design and Installation Considerations for ESPs

High Volume Lift Capacities	- Excellent. - Limited by needed HP and can be restricted by casing size.
Low Volume Lift Capacities	- Generally poor. - Lower efficiencies and high operating costs <400 BFPD.
High viscosity fluid handling Capacities	- Fair. Limited to as high as 1000 cp. - Depends on economics. (~>7-9°API) - Increases HP and reduces head.
Depth limits	- Usually limited to motor HP or temperature. - Practical depth about 10,000 feet. - 1000' – 10000' TVD is typical, Max is 15,000 ' TVD

ESP Design

Input

Tubing OD	3.5	inches
Tubing ID	2.992	inches
Top Perforation	6571.52	ft
Pump Intake Depth	5620	ft
z factor	0.85	
Oil Gravity	35	API
Water Cut	90	%
Water Gravity	1.01	rel-H2O
Gas Gravity	0.8	rel-air
Producing GOR	200	scf/stb
Bubble Pressure	964	psi
Bottomhole Temperature	90	°C
Wellhead Temperature	50	°C
Wellhead Pressure	87	psi
Casing-Head Pressure	15	psi
Reservoir Pressure	2320.6	psi
Desired production rate	3500	bfpd
Pump Intake Pressure	1160	psi
Present Production Rate	2500	bfpd
	158	
Correction factor	120	

Output

Step 1 Production Capacity			
	S.G (Oil)	0.85	
	SG (Water)	0.994	
	Flowing pressure (Pwf)	1570	psi
	Productivity Index (PI)	3.329	bpd/psi
	Pwf @ Desired Flow rate	1269	psi
	Pump Intake pressure	853.6	psi
Step 2 Gas Calculation			
	Solution GOR (Rs)	190.5	scf/stb
	Oil Formation Volume Factor (Bo)	1.131	bbl/stb
	Gas Formation Volume Factor (Bg)	3.086	bbl/Mscf
	Total Gas Volume	70	Mscf
	Solution gas volume	66.67	Mscf
	Free Gas Volume	3.328	Mscf
	Volume of oil at the pump intake (Vo)	395.9	bopd
	Volume of gas (Vg) at the pump intake	10.27	bgpd
	Volume of water (Vw) at the pump intake	3150	bwpd
	Total Fluid Volume	3556	bfpd
	Free Gas %	0.289	%
	Total Mass of Produced Fluid (TMPF)	1E+06	
	SGcomp	0.997	
Step 3 Total Dynamic Head (TDH)			
	PIP	1990	ft
	Net Well Lift	3630	ft
	well-tubing friction loss	218.3	ft
	Wellhead Pressure Head (Hwh)	201.4	ft
	Total Dynamic Head	4049	ft

ESP Design – Stages Required

From the manufacturer's catalog information, select the pump type which will be operating within the recommended operating range and with the highest efficiency at the calculated capacity 3556 bfpd (565 m³/d).

The expected production profile of the well has to be covered by the operating range of the pump during the expected ESP run life.

For this example, Select the 538 series G31 pump

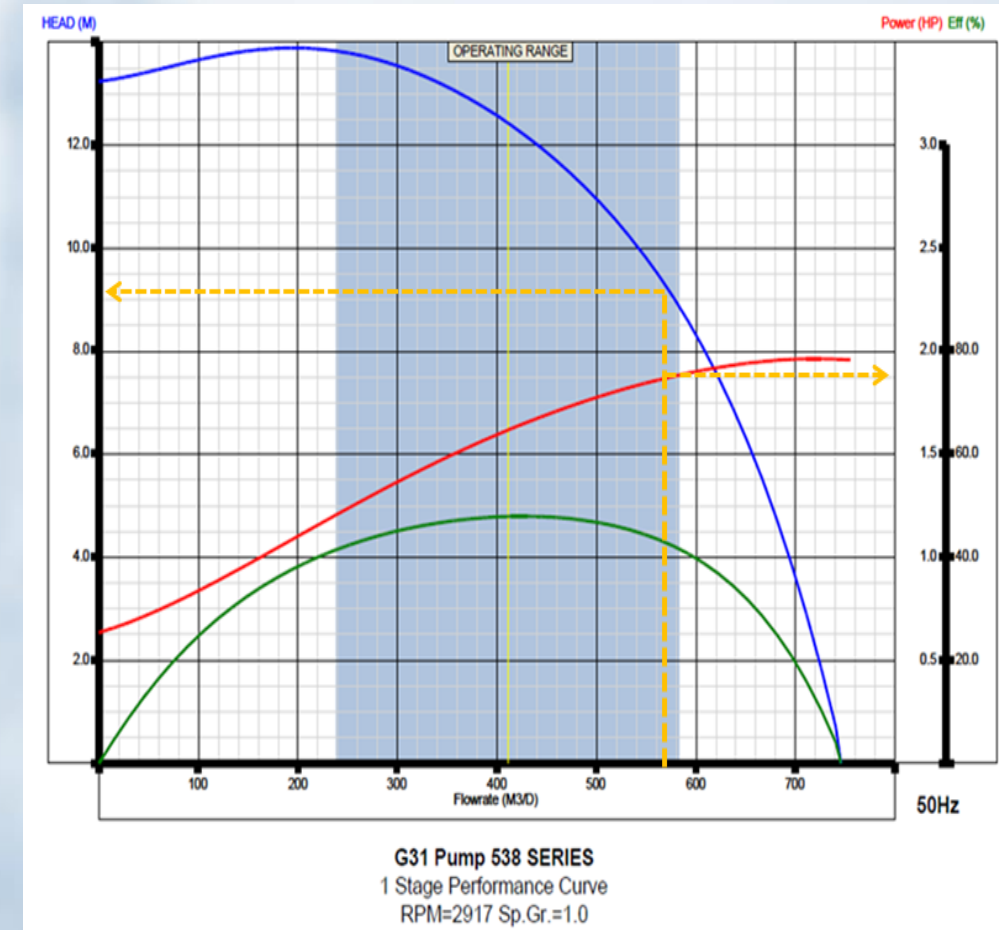
From the selected pump performance curve, The head in meters for one stage at 565 m³/d (3556 bfpd) is **8.6 m (28.2 ft.)**.

The BHP per stage is **1.875 hp**.

Total number of stages required = TDH/ (head/stage)

Total number of stages required = (4049 /28.1) = 143 stages.

Stages required = 143 stages.



ESP Design - Total BHP Requirement

Pump BHP required:

Once the required number of pump stages is decided, calculate the Pump BHP required:

Pump BHP required = BHP/stage $\times$ number of stages $\times$ SGcomp

Pump BHP required = $1.875 \times 143 \times 0.997 = 267.32$ hp

Pump BHP required = 267.32 hp

From the manufacturer's catalog, the 513 series GSB3 DB HL (double-bag, high load thrust bearing) [3 chambers, Bag/Bag/Labyrinth] seal section is selected for this well .

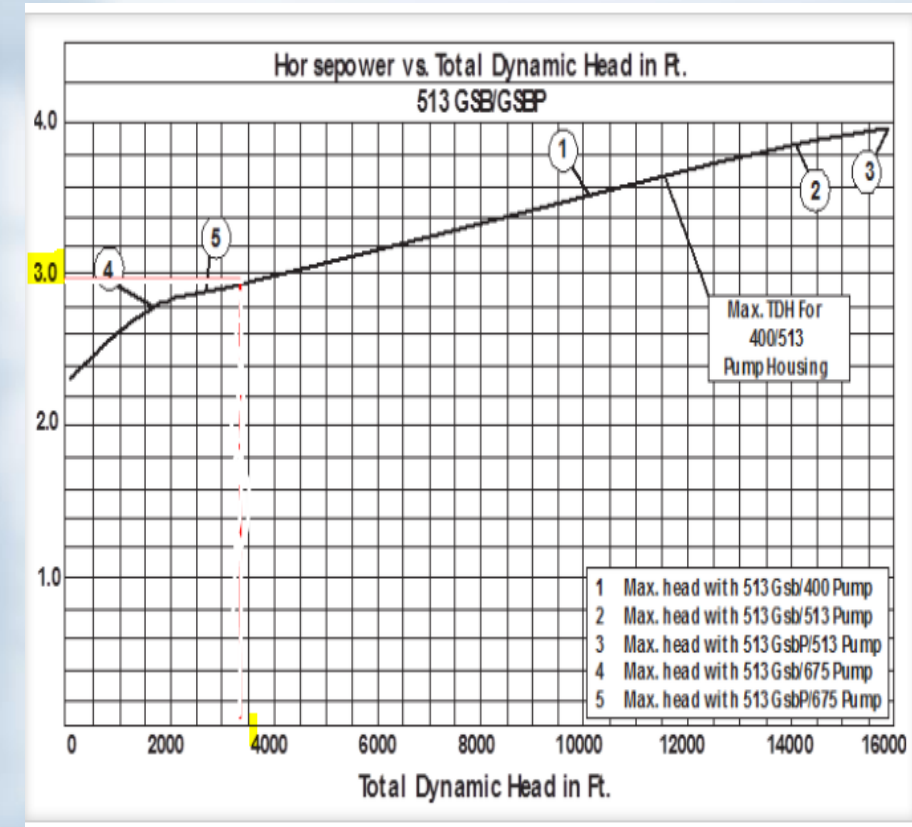
From the Chart, HP required for the seal is **3 hp**

Therefore:

Total HP requirement = Pump BHP requirement + Seal HP requirement

Total HP requirement = $267.32 + 3$

Total HP requirement = 270.32 hp



ESP Design - Motor and Cable Selection

For this example, 333 hp 562 series motor (**333 hp; 2321 V; 88 amps**) seems to be a good choice.

Typically, the motor is selected to operate in the range from 70 to 100% of its rating.

Determine Cable Size:

Cables #1 and #2 have **voltage drop of 26 volts/1,000 ft. and 20 volt/1000 ft. @ 25°C (77°F)** respectively. Calculate their voltage drop at the downhole temperature (158°F in our case).

For cables operating at a different temperature, the voltage drop can be determined by multiplying it by Temperature Correction Factor (TCF)

$$TCF = 1 + 0.00214 \times (T - 77)$$

$$TCF = 1 + 0.00214 \times (158 - 77) = 1.173$$

Therefore:

cable #1 conductor voltage drop @ (T=158°F) = $20 \times 1.173 = 23.46$ volts/1,000 ft.

cable #2 conductor voltage drop @ (T=158°F) = $26 \times 1.173 = 30.5$ volts/1,000 ft.

As low voltage drop is required, select **Cable #1**

